

Predicting Flight Delays and Their Causes Across Major U.S. Airports

Exploratory Data Analysis

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<https://github.com/yalharbii/flight-delay-eda>

Summary of Research Questions

1. Which factors, including airport, airline, scheduled time, and reported delay cause, are most associated with longer flight delays?
2. Which airports in the available extract have the longest average departure and arrival delays?
3. Does the reported weather-delay category identify flights that are more likely to have departure delays longer than 30 minutes?

Motivation

Flights are frequently delayed, and millions of people experience this phenomenon yearly. However, besides weather conditions, there are various reasons for flights to be delayed. These reasons include airport congestion, airline management, the time at which one is scheduled to leave, among others. Being passionate about aviation and pursuing a pilot certificate, I would like to know more about the main reasons for flight delays and see if it is possible to predict them. This will give me an opportunity to make better decisions regarding traveling and gain an understanding of how various factors affect airline performance.

Dataset

For the EDA, a 77-row exploratory extract of a publicly available flight delay dataset in 2015 of the U.S., generated from Bureau of Transportation Statistics On-Time Performance Records is used. Each row represents one flight record, including its origin airport, destination airport, scheduled departure time, delay outcomes, and reported delay cause. The extract has 17 original variables such as date components, carrier, airport attributes (origin and destination), scheduled departure, departure and arrival delays, distance, cancelled flag, reported reason of delay, and airport attributes. Three engineered variables (departure hour, time-of-day, and 30+ minutes of departure delay indicator) were created in the process of analysis.

The extract includes Seattle-Tacoma International (SEA), Dallas/Fort Worth International (DFW), and Chicago O'Hare International (ORD). Attributes of the airports are included in the extract along with the flight records. The original plan included a merge of the NOAA weather dataset; as the observations of the NOAA dataset were not provided in the submitted data, this EDA uses the Weather Delay category rather than the precipitation, visibility, temperature, or wind measures. Thus, the objective of multiple datasets is partly achieved in this case.

Method

The Python code first imports the data using pandas, generates the Boolean variable DELAY_OVER_30, transforms the scheduled departure times to hours, and categorizes the hours as overnight, morning, afternoon, evening, and late night. Missing data is determined using DataFrame.isna().sum(). Quantitative variables are analyzed using DataFrame.describe() function, while the categorical variables are analyzed using value_counts(). The next step is to classify the data based on destination airports and delay reasons. There are three plots generated using matplotlib for the following comparison: average airport delay, counts of delay reasons, and percent delayed more than 30 minutes.

EDA Results

Size, Rows, and Columns

The raw data contains 77 rows and 17 columns. Following the feature engineering phase, the analysis table consists of 20 columns. A row corresponds to a 2015 flight record, whereas the columns contain information on the date, the airline, route, timings, delay results, distances, reason for delay, and airport characteristics of the destination airport.

Missing Data

The Pythonic missing value check shows two missing ARRIVAL_DELAY values, representing 2.60% of the records. All other columns have no missing values. The missing arrival delays are kept in datasets for those analysis which don't involve arrival delays and are automatically ignored by pandas in calculation of the average arrival delay. In any future model, one should either remove those records for arrival delay result or investigate if they correspond to cancellations or incomplete data.

Variables of Interest and Summaries

The primary quantitative variables include DEPARTURE_DELAY, ARRIVAL_DELAY, and DISTANCE. While the first two refer to how many minutes the flight was delayed, the third variable shows how long the route was, measured in miles. The categorical variables are IATA_Code, AIRLINE, CANCELLATION_REASON, and TIME_OF_DAY. The use of these variables is justified by the research questions since they enable comparisons between different airports, airlines, reasons for delays stated, and times of departure. The Boolean variable is DELAY_OVER_30, which indicates if there was a delay of more than 30 minutes.

Variable	Mean	SD	Min	Q1	Median	Q3	Max
Departure delay	40.47	25.76	16	23	34	41	100
Arrival delay	22.79	32.04	0	5	13	23.5	171
Distance	581.22	503.59	67	247	491	802	3711

Categorical counts: Carrier Delay = 44 (57.1%), National Air System Delay = 16 (20.8%), Weather Delay = 15 (19.5%), and Late Aircraft Delay = 2 (2.6%). Airline counts were MQ = 42, AA = 13, OO = 13, AS = 4, EV = 3, DL = 1, and NK = 1. These counts show that the extract is not balanced across airlines, so airline comparisons would not yet be reliable.

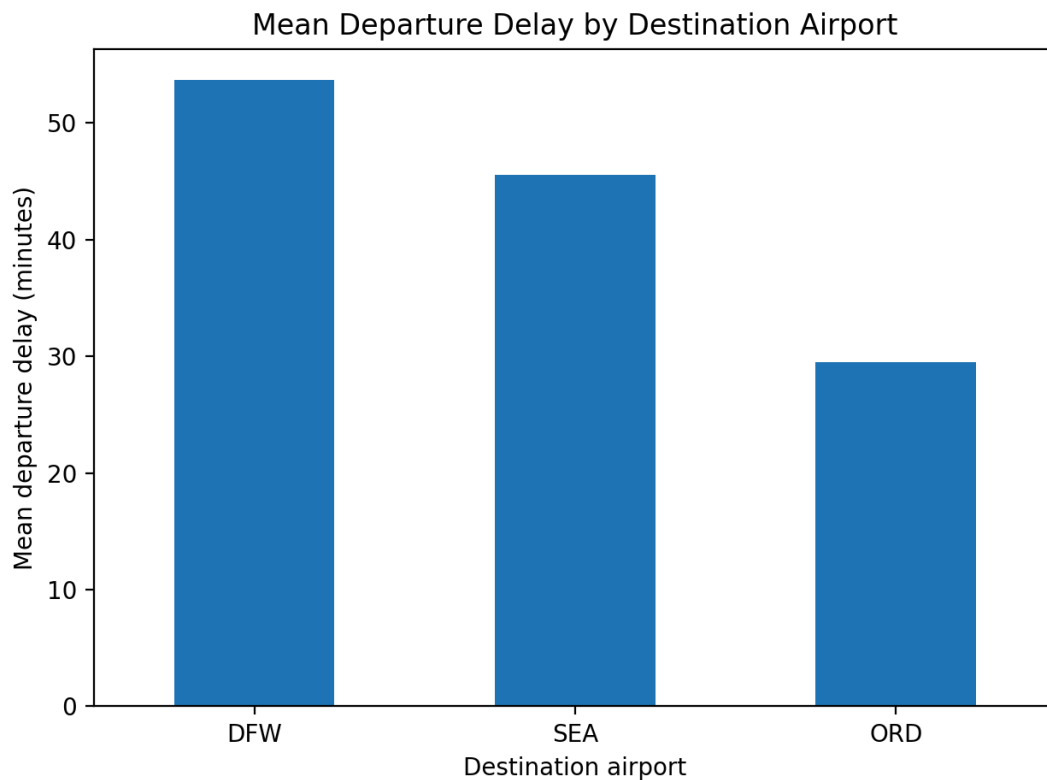


Figure 1. DFW has the highest mean departure delay in this extract, followed by SEA and ORD.

The following is a bar graph that illustrates the mean departure delays at the three destination airports. Bar graphs are appropriate since airport type is categorical data while the result is the mean of a group of observations. The airport DFW had the highest mean departure delay of 53.68 minutes. SEA had a mean of 45.53 minutes while ORD had a mean of 29.49 minutes. 72.0%, 80.0% and 48.6% were the percentages delayed beyond 30 minutes at DFW, SEA and ORD, respectively.

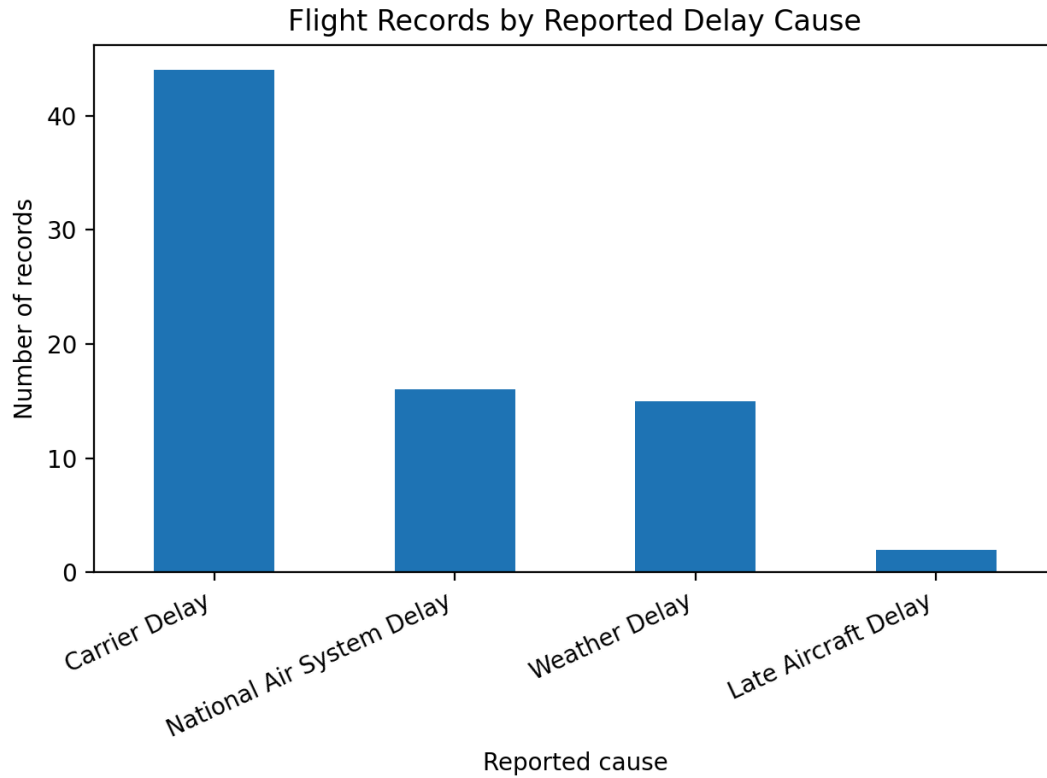


Figure 2. Carrier-related records are the most common, while weather accounts for about one-fifth of the extract.

The second graph reveals how many instances each of the recorded delay causes has occurred. As seen from the plot, carrier delay has been registered 44 times, national air system delay – 16, and weather delay – 15 times. That is in line with the rationale behind the analysis, that delay causes are not limited to the weather factor only. However, this classification does not represent any weather data directly.

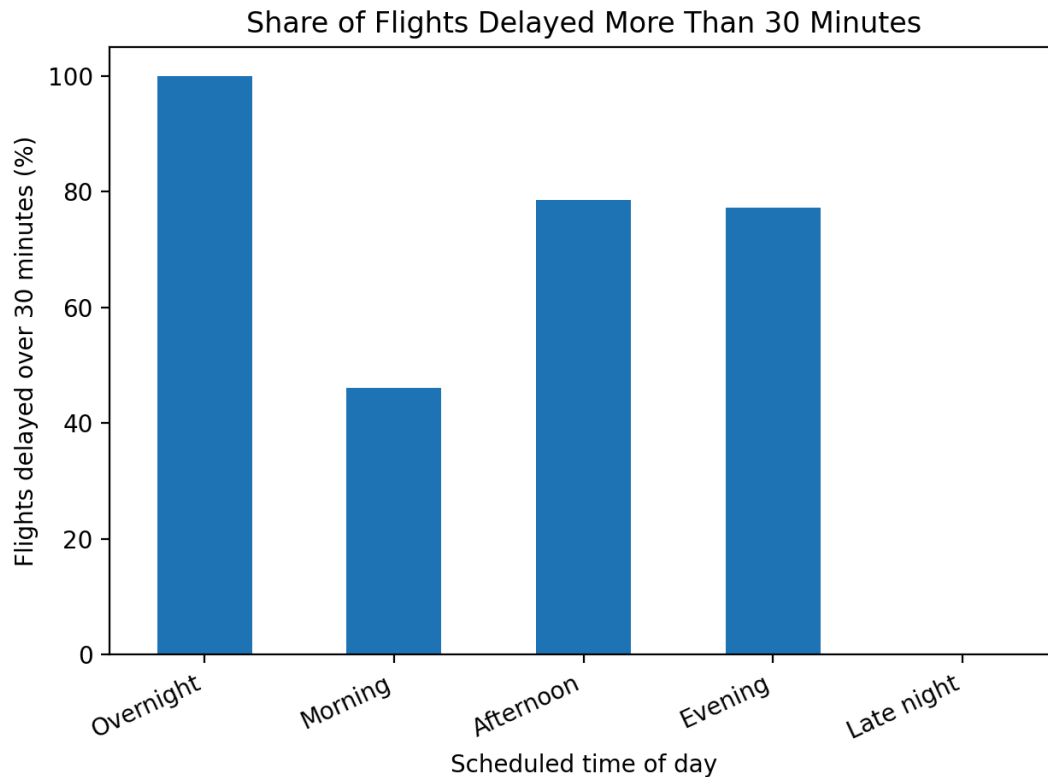


Figure 3. Afternoon and evening flights have higher over-30-minute delay shares than morning flights in the extract.

The third graph shows the percentages of the number of records of departure delays greater than 30 minutes in the scheduled time-of-day intervals. For afternoon and evening time periods, these percentages were 78.6% and 77.3%, respectively, while in the morning period, only 46.2% of the records were recorded. As for the overnight period, there were just two observations, which leads to a 100% percentage.

Challenge Goals Update

Multiple data sets for the EDA have been downgraded. Airport data has been available as pre-joined data, but NOAA weather data has not been joined yet. The next step would be to get hourly/daily NOAA weather data and then join on the basis of airport and time. Machine learning is still going ahead as per plan. Logistic regression, random forest, and gradient boosting algorithms can be compared once we get the complete data set. Delay over 30 minutes would be our target variable.

Work Plan Evaluation

The amount of time needed for finding datasets to match and creating join keys was underestimated. The time taken by cleaning the flight data and preparing summary statistics and plots was about as planned.

The process of integration of the weather data from the NOAA observations, however, took more time since the identification numbers of weather stations do not match the codes of the airports and flight departure time.

The following tasks include acquiring the full flight dataset of the BTS, matching the NOAA weather stations and the airports, checking and cleaning the merged dataset, performing an extended exploratory analysis and preparing the classification models. Data acquisition, mapping of stations, merging of data, validation and cleaning will take approximately 5-7 hours. Additional time of 2-3 hours may be required for additional exploration and tests. Creation and comparison of machine learning models will require 3-4 hours, while another 2 hours will be required for report revision and cleaning of the repository.

Testing

I verified the EDA functions using a different Python file named `test_eda.py`. This includes the verification of loading of the dataset, presence of required columns, correct calculation of missing value summaries, presence of correct statistics in quantitative summaries, and correct grouping of airports. I also executed `eda.py` from start to finish and checked if there are any errors in the creation of summary tables and visualizations. Additionally, I also used `flake8` to check for any formatting problems in both Python files and fixed them.

Collaboration

These resources are included: assignment requirements, pandas documentation, matplotlib documentation, description of Bureau of Transportation Statistics On-time Performance data, and publicly available 2015 flight delays extract, which was used for performing this EDA.

Data limitations: This submission represents a completely reproducible draft of EDA performed using a small public extract. All conclusions are exploratory and need to be revised after replacement of `flight_sample.csv` with the whole project datasets and execution of `eda.py`.